

• femora

WOMEN'S HEALTH • ON-DEVICE AI • URDU & ENGLISH

Every cycle, every risk, every result — explained to her, **in her own language.**

Femora brings cycle prediction, PCOS screening, breast-health AI and a generative-AI companion into one app that runs on her phone, in the language she actually speaks.

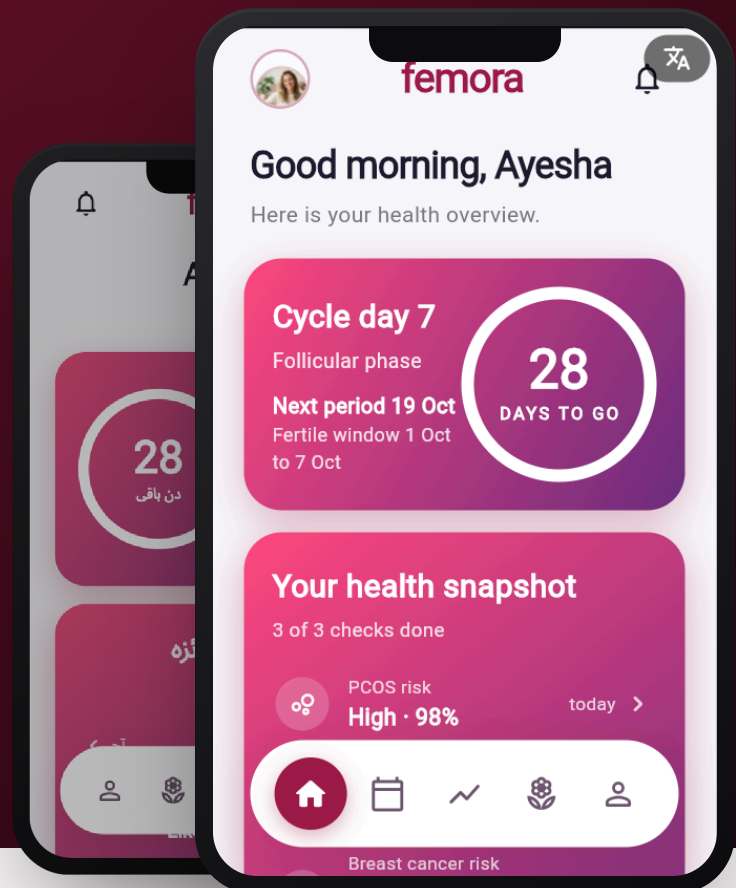
Cycle & fertility prediction

PCOS risk + what-if simulator

Breast ultrasound AI

Bilingual AI companion

Photographed report reader



THE GAP THIS FILLS

Two apps and a browser tab weren't enough.

Most women in Pakistan piece their care together from a generic period tracker, a risk calculator built for someone else's population, and a search bar for anything a doctor hands them on paper.

Before Femora

- ✕ A cycle tracker that has never heard of PCOS or breast health
- ✕ Risk calculators trained on someone else's population, with no stated limits
- ✕ Every screen only in English
- ✕ A photographed lab report means a search engine, not an explanation
- ✕ Her results scattered across other people's clouds

With Femora

- ✓ Cycle, PCOS, breast health and an AI companion in one on-device app
- ✓ Screening models trained and measured, with every limit stated plainly
- ✓ Every screen bilingual — Urdu and English, automatically right to left
- ✓ Photograph any report and have it explained, not just translated



Results stay on her phone unless she chooses to share them

01 • Cycle & fertility

Predictions that get better every time she logs one.

The estimate blends her own history with a measured population baseline, so it starts useful from day one and sharpens with every cycle she tracks.

1 Personal + population blend

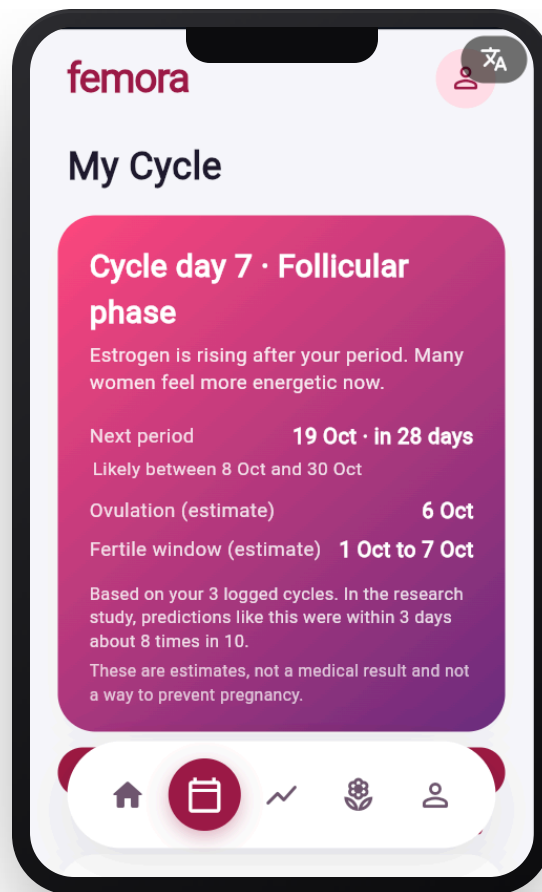
Not a rigid 28-day guess — her own average, weighted more heavily the more she logs.

2 Phase, ovulation & fertile window

Calendar view with the current phase, next period and fertile window at a glance.

3 Backtested, not assumed

The dashboard replays the prediction against her own logged cycles to show how it actually did.



02 · PCOS screening

A risk score she can actually act on.

The questionnaire is scored by a gradient-boosted model trained with monotonic constraints, then handed back as something she can explore, not just read once.

1 Direction-safe by construction

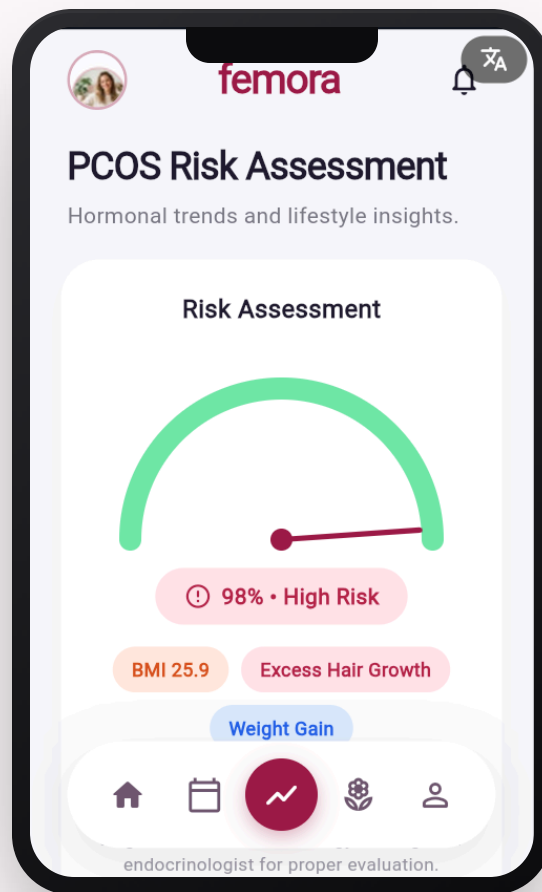
A symptom can only ever push the estimate the clinically consistent way — verified by an automated 128-case test.

2 Live what-if simulator

Move a slider for exercise, diet or weight and watch the model's estimate update in real time.

3 Explained, not just scored

Per-answer SHAP attributions surface the main factors behind her result.



03 • Breast health

An ultrasound second opinion, and a guide for the exam itself.

A ResNet50 classifier reads an uploaded ultrasound scan, a risk questionnaire scores her background against 2.39 million real screening records, and an illustrated guide walks her through a monthly self-exam.

1 Trained across four hospitals

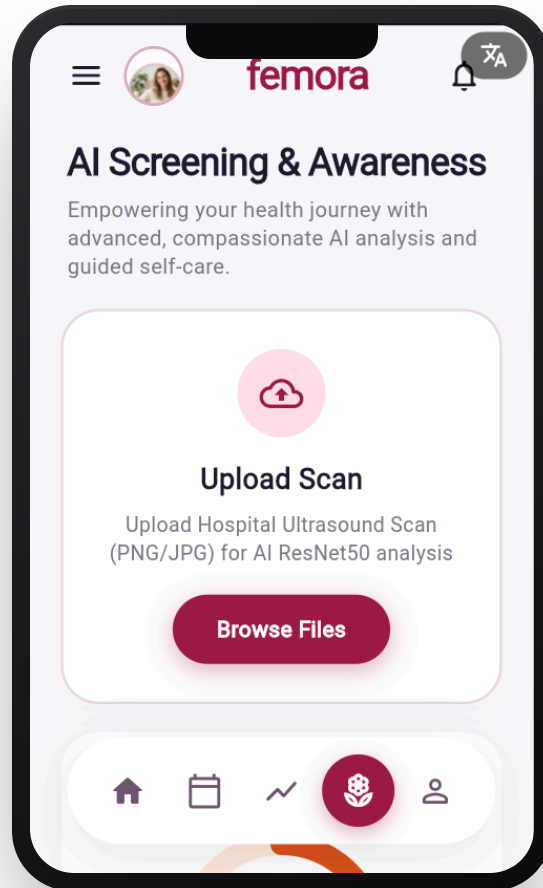
Combining scanner sources raised the AUC on scans it had never seen from 0.69 to 0.87.

2 A gate that knows what it doesn't know

Non-breast ultrasounds are recognised and refused before they reach the classifier.

3 Guide, log and reminder together

The self-exam guide, the logging and the monthly reminder live in the same tab as the screening tools.



04 • AI companion

Ask anything, in Urdu or English, out loud or typed.

A generative-AI companion that already knows her own results — without ever knowing her name — and speaks back in whichever language she asked in.

1 Personal, never identified

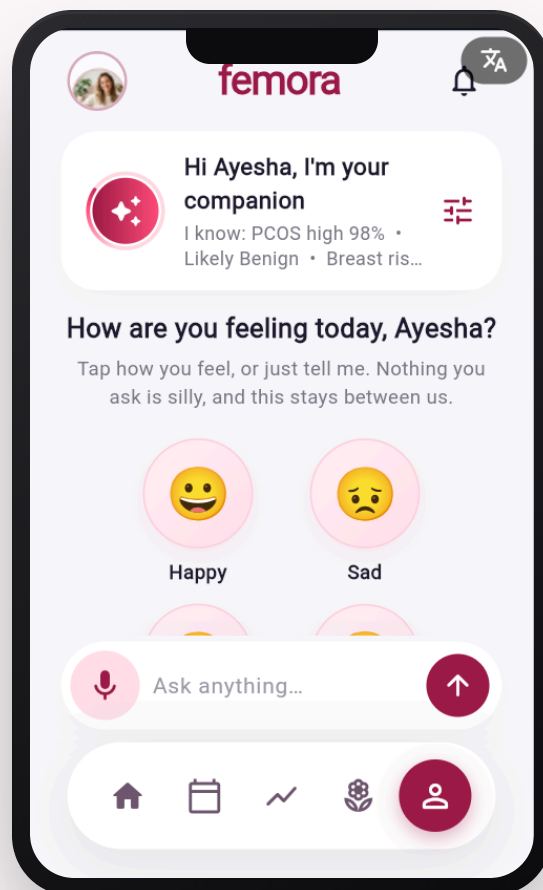
A name-free summary of her results is all the model ever receives.

2 Voice in, voice out

Speak or type the question; hear the answer read back in Urdu or English.

3 Safety enforced in code

Emergency wording is caught by server-side rules, not left to the model's judgement.



05 • Report reader

Photograph a lab report. Get it explained, not just translated.

A photo of a blood test, hormone panel, ultrasound report or prescription comes back as a structured, plain-language explanation, in whichever language she reads best.

1

Never invents a number

Only values actually printed on the report are repeated back; a printed range is always used over a remembered one.

2

Escalates on its own

A single critical value forces an urgent flag, whatever else the model returned.

3

Identifiers scrubbed automatically

Names, CNIC numbers and phone numbers are removed from the answer before she ever sees it.

06 • BUILT FOR PAKISTAN

The whole app, mirrored — not just translated.

Switching to Urdu doesn't swap a few labels: the entire layout flips right to left, automatically, and every button, form and dialogue reads in Urdu, in a proper Nastaliq face.



ENGLISH • LEFT TO RIGHT

اردو • RIGHT TO LEFT

MEASURED, NOT ASSUMED

Every number here comes from a held-out test set.

0.87

Ultrasound AUC on scans
from a hospital never seen
in training

84.4%

PCOS screening accuracy,
AUC 0.88

2.39M

Real screening
mammograms behind the
breast-risk model

350+97

Automated app & backend
tests, all passing

See the code, or run it yourself.

Femora is a final-year project at Air University Islamabad — open on GitHub, with every measured result reproducible from the repository.

Femora

Cycle, PCOS, breast health & an AI companion — one on-device app.

REPOSITORY

github.com/ammad-sajjad/Femora

BUILT BY

Ammad Sajjad

PROGRAMME

BS — Air University Islamabad

[View the repository](#)

Femora · Final-year project · Air University Islamabad